

Deloitte Data Analytics Virtual Internship(Forage)

Macaro Industries

Here is the background information on your task

Using a data unification algorithm, the tech team at our client, Daikibo, has converted all telemetry data collected from its 4 factories:

- Daikibo Factory Meiyo (Tokyo, Japan)
- Daikibo Factory Seiko (Osaka, Japan)
- Daikibo Berlin (Berlin, Germany)
- Daikibo Shenzhen (Shenzhen, China)

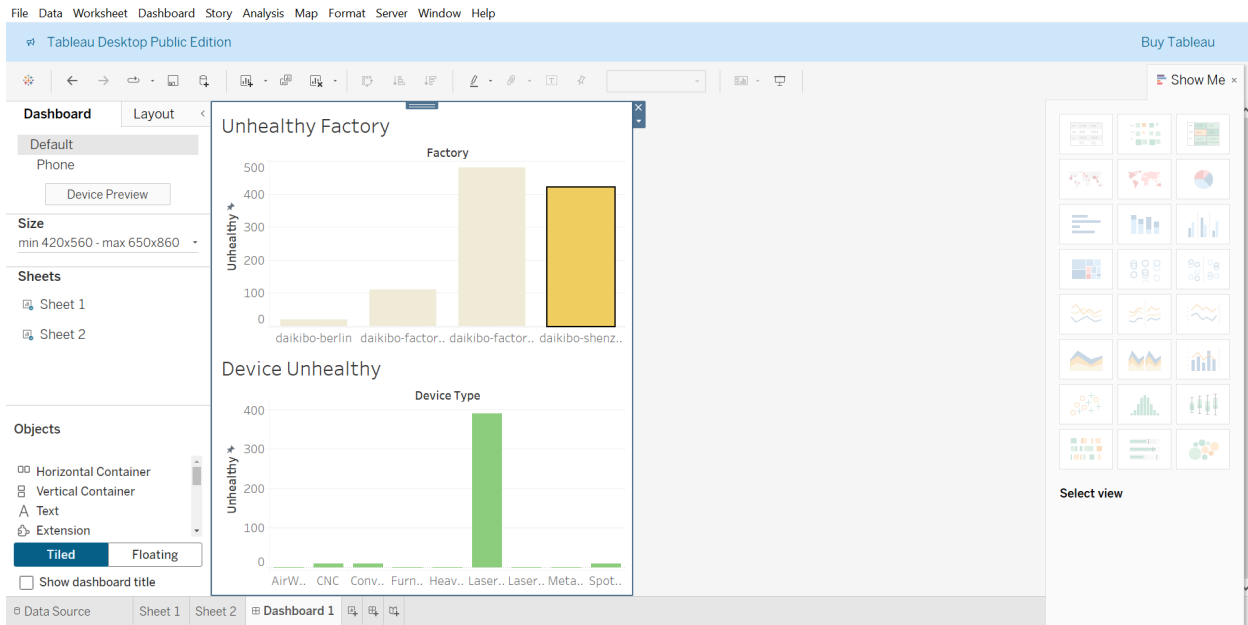
Each location has 9 types of machines, sending a message every 10 mins. Daikibo has been collecting this data for one month (May 2021) and they've shared this data in the form of a single JSON file.

The reason the client wanted to collect telemetry was to answer 2 questions:

1. In which location did machines break the most?
2. What are the machines that broke most often in that location?

Dashboard Link:

<https://public.tableau.com/app/profile/ramesh.k2258/viz/MacaroIndustriesDashboard/Dashboard1?publish=yes>



Here is your task

Your task is to analyse the telemetry data collected by Daikibo in a software called Tableau. Here are the steps that you need to take:

1. Download the free trial of Tableau (link in the Resources).
2. Install Tableau on your computer and register an account with the same email you used to download the software.
3. Download the daikibo-telemetry-data.json.zip file -> unzip -> and import it in Tableau.
4. Create a calculated measure field called "Unhealthy" with a value of 10 for every unhealthy status (representing 10 mins of potential down time since the previous message).
5. Create a bar chart called "Down Time per Factory".
6. Create a new sheet with a new bar chart called "Down Time per Device Type".
7. Create a Dashboard with the 2 previous sheets and set the first chart to be used as a filter (selecting a factory in the first chart shows only the down time of the machines in this factory in the second chart).
8. Select the factory with the most down time (click on its bar), make a screenshot of the dashboard and upload it as a submission for this task.

Here is the background information on your task

After a worrisome number of internal complaints about gender inequality in terms of salary, Daikibo Industrials wants us to help them investigate.

The Forensic Tech team has built an algorithm to quantify “level of gender pay equality” for most job roles within the company, in all company locations. Our Forensics lead thinks it would be a great idea for you to finish the job.

	A	B	C	D
1	Factory	Job Role	Equality Score	Equality class
2	Daikibo Factory Meiyo	C-Level	-25	Highly Discriminative
3	Daikibo Factory Meiyo	VP	-26	Highly Discriminative
4	Daikibo Factory Meiyo	Director	-19	Unfair
5	Daikibo Factory Meiyo	Sr. Manager	-15	Unfair
6	Daikibo Factory Meiyo	Manager	-14	Unfair
7	Daikibo Factory Meiyo	Jr. Manager	-20	Unfair
8	Daikibo Factory Meiyo	Sr. Engineer	-5	Fair
9	Daikibo Factory Meiyo	Engineer	-8	Fair
10	Daikibo Factory Meiyo	Jr. Engineer	3	Fair
11	Daikibo Factory Meiyo	Operational Support	-22	Highly Discriminative
12	Daikibo Factory Meiyo	Machine Operator	-7	Fair
13	Daikibo Factory Seiko	VP	-19	Unfair
14	Daikibo Factory Seiko	Director	-10	Fair
15	Daikibo Factory Seiko	Sr. Manager	-21	Highly Discriminative
16	Daikibo Factory Seiko	Manager	-21	Highly Discriminative
17	Daikibo Factory Seiko	Jr. Manager	-24	Highly Discriminative
18	Daikibo Factory Seiko	Sr. Engineer	-4	Fair
19	Daikibo Factory Seiko	Engineer	-7	Fair
20	Daikibo Factory Seiko	Jr. Engineer	4	Fair
21	Daikibo Factory Seiko	Operational Support	-19	Unfair
22	Daikibo Factory Seiko	Machine Operator	-5	Fair
23	Daikibo Berlin	Sr. Manager	-15	Unfair
24	Daikibo Berlin	Manager	-16	Unfair
25	Daikibo Berlin	Jr. Manager	-17	Unfair
26	Daikibo Berlin	Sr. Engineer	4	Fair
27	Daikibo Berlin	Engineer	2	Fair
28	Daikibo Berlin	Jr. Engineer	4	Fair
29	Daikibo Berlin	Operational Support	0	Fair